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$$\begin{aligned} \text{i)} \quad & \| \vec{u} + \vec{v} \|^2 - \| \vec{u} - \vec{v} \|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle = \\ & = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) \\ & = 2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle = 2\langle \vec{u}, \vec{v} \rangle + 2\overline{\langle \vec{u}, \vec{v} \rangle} \end{aligned}$$

↳ skoro symetrie $\langle \cdot, \cdot \rangle$.

Pro $x \in \mathbb{C}$ platí $x + \bar{x} = 2\operatorname{Re}(x)$, 2

$$\begin{aligned} 2\langle \vec{u}, \vec{v} \rangle + 2\overline{\langle \vec{u}, \vec{v} \rangle} &= 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \| \vec{u} + \vec{v} \|^2 - \| \vec{u} - \vec{v} \|^2 \\ \Rightarrow \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) &= \frac{1}{4} \cdot (\| \vec{u} + \vec{v} \|^2 - \| \vec{u} - \vec{v} \|^2), \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad & \| \vec{u} - i\vec{v} \|^2 - \| \vec{u} + i\vec{v} \|^2 = \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle \\ & = \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle \\ & + \langle \vec{v}, \vec{v} \rangle) = 2i\langle \vec{v}, \vec{u} \rangle - 2i\langle \vec{u}, \vec{v} \rangle = 2i(\overline{\langle \vec{u}, \vec{v} \rangle} - \langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

* pro $x \in \mathbb{C}$ platí: $\bar{x} - x = -2i\operatorname{Im}(x)$

$$\begin{aligned} \Rightarrow 2i(\overline{\langle \vec{u}, \vec{v} \rangle} - \langle \vec{u}, \vec{v} \rangle) &= 2i(-2i\operatorname{Im}\langle \vec{u}, \vec{v} \rangle) = 4\operatorname{Im}\langle \vec{u}, \vec{v} \rangle \\ \Rightarrow \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) &= \frac{1}{4} (\| \vec{u} - i\vec{v} \|^2 - \| \vec{u} + i\vec{v} \|^2), \end{aligned}$$

Použili jsme axiom skoro symetrie ($\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$),
a obě axiom linearity ($\langle i\vec{u}, \vec{v} \rangle = i\langle \vec{u}, \vec{v} \rangle$, $\langle i\vec{u}, i\vec{v} \rangle = -i\langle \vec{u}, \vec{v} \rangle$,
skalár ^{části} $t \in \mathbb{R}$ je lineární ve druhé složce a konjugovaně
lineární ve první složce.